

TOPIC NAME: Lecture 16 (ROC)

DAY: _____

Azmari Sultana

TIME: _____

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When there is class imbalance, we can't use

$$\text{Accuracy} = \frac{\text{correctly predicted}}{\text{Total prediction}}$$

In this case, we need to use a different evaluation matrix. That is Confusion Matrix.

Gender	Age	Chest Pain	Blood Press	Blood Sugar	Actual Target	Predicted Target
M	42	3	150	1	0	1
M	55	2	120	0	0	0
F	46	1	145	1	0	0
M	38	0	160	1	1	0
F	49	2	140	0	0	0
F	44	3	130	1	1	1
M	58	0	111	0	0	0
M	52	1	120	0	0	0
M	46	3	150	1	0	0
F	36	2	133	1	1	0
M	60	3	140	1	0	0
F	58	4	145	0	0	0

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	1 = cancer	0 = Not cancer
1 = cancer	1 TP	2 FN
0 = no cancer	1 FP	8 TN

Column = predicted target

Row = actual target

True Positive (TP): Number of correctly identified positive cases

False Negative (FN): Number of cases where actual Positive cases are identified as Negative.

True Negative (TN): Number of correctly identified Negative cases.

False Positive (FP): Number of cases where actual Negative cases are identified as Positive.

Consideration: Where the FN & FP are less in number

Sensitivity And Specificity

Sensitivity = From "Actual True" values how many are predicted as "True"

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{1}{1+2} = 0.33 = 33\%$$

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Specificity = From "Actual False" values how many are predicted as "False"

$$\text{Specificity} = \frac{TN}{FP + TN}$$

$$= \frac{8}{1+8} = \frac{8}{9} = 0.889 = 88.89\%$$

Recall And Precision:

Precision = From "Predicted True" values how many are actually "True".

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{1}{1+1} = 50\%$$

Recall = From "Actual True" values how many are predicted as "True".

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$= \frac{1}{1+2} = 33\%$$

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		Predicted class		
		Yes	No	
Actual class	Yes	2750 (TP)	250 (FN)	3000
	No	150 (FP)	2850 (TN)	3000
		2900	3100	6000

Confusion Matrix

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{2750}{2750 + 250}$$

$$\text{Specificity} = \frac{TN}{FP + TN} = \frac{2850}{150 + 2850}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{2750}{2750 + 150}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{2750}{2750 + 250}$$

		Predicted class		
		class A	class B	class C
Actual class	class A	2900 (TP _A)	75	25
	class B	100	800 (TP _B)	100
	class C	50	100	850 (TP _C)

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class A

$$\text{Sensitivity (class A)} = \frac{TP}{TP+FN} = \frac{900}{900+(75+25)}$$

$$\begin{aligned} \text{Specificity (class A)} &= \frac{TN}{FP+TN} = \frac{800+100+100+850}{(100+50)+(800+100+100+850)} \\ &= \frac{1850}{2000} \end{aligned}$$

$$\text{Precision (class A)} = \frac{TP}{TP+FP} = \frac{900}{900+(100+50)} = \frac{900}{1050}$$

$$\text{Recall (class A)} = \frac{TP}{TP+FN} = \frac{900}{900+(75+25)} = \frac{900}{1000}$$

class B

$$\text{Sensitivity / Recall} = \frac{TP}{TP+FN} = \frac{800}{800+(100+100)} = \frac{800}{1000}$$

$$\begin{aligned} \text{Specificity} &= \frac{TN}{FP+TN} = \frac{900+25+50+850}{(100+75)+(900+25+50+850)} \\ &= \frac{1825}{2000} \end{aligned}$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{800}{800+(100+75)} = \frac{800}{975}$$

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class c

$$\text{Sensitivity / Recall} = \frac{TP}{TP + FN} = \frac{850}{850 + (100 + 50)} = \frac{850}{1000}$$

$$\begin{aligned} \text{Specificity} &= \frac{TN}{FP + TN} = \frac{900 + 75 + 100 + 800}{(100 + 25) + (900 + 75 + 100 + 800)} \\ &= \frac{1875}{2000} \end{aligned}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{850}{850 + (100 + 25)} = \frac{850}{975}$$

ROC AUC

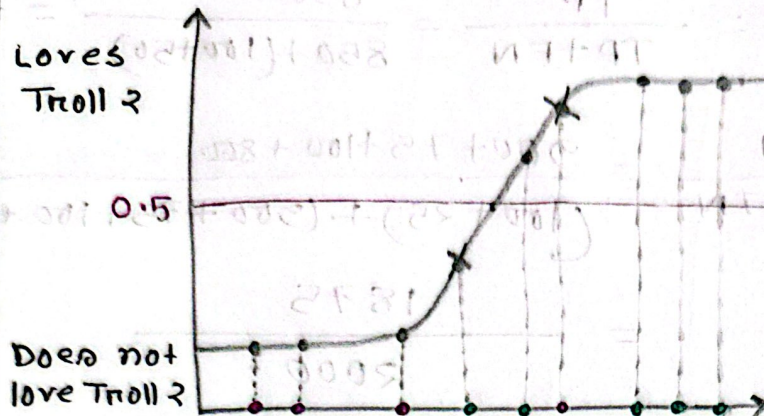
Popconn	Love Troll 2	Prediction	Predicted target
3	0	0.2	0
4	0	0.2	0
7	0	0.4	0
9	1	0.4	0
11	1	0.6	1
12	0	0.6	1
16	1	0.7	1
17	1	0.7	1
18	1	0.8	1

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Predicted target

1 = loves troll? 0 = Does not love troll?

1 = loves troll?

0 = Does not loves troll?

4	1
1	3

Actual target

Here, if we increase or decrease the threshold, the number inside the matrix will change. So, in this case, it is tough to know the correct threshold. So, we need to use ROC curve in this case.

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ROC curve = Receiver Operating Characteristic Curve

$$\text{True Positive Rate (TPR)} = \frac{\text{TP}}{\text{Actual positive}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

↓
Sensitivity

$$\text{False Positive Rate (FPR)} = \frac{\text{FP}}{\text{Actual Negative}} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$

↓
1 - Specificity

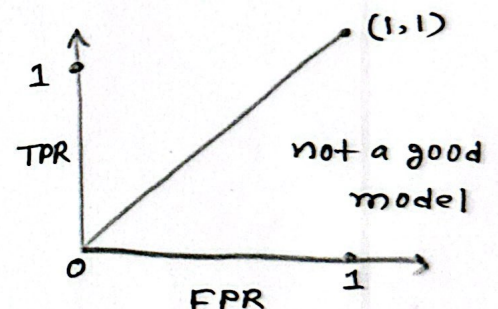
		Predicted	
		1	0
Actual	1	4	0
	0	4	0
		TP	FN
		FP	TN

$$\text{TPR} = \frac{4}{4+0} = 1$$

$$\text{FPR} = \frac{4}{4+0} = 1$$

We need: TPR ↑ FPR ↓

TPR = FPR = 1 → not desirable



So in the ROC curve, where TPR ↑ FPR ↓, we would know our desirable point there.

AUC → Area Under the curve.

From ROC curve, we can calculate the AUC. Whoever have the most AUC in number, that model is best.